

Supplementary Material for CauSample

ANONYMOUS AUTHOR(S)

A Proof of the Causal Sampling Bound

We prove the per-child recovery and conditional bound comparison in Theorem 4.2. The probability in its first claim is for an individual child given exact parent trajectories. We then give a separate whole-batch extension, including the root recovery conditions and its simultaneous-probability allocation.

Notation and assumptions. Fix a batch $X_i \in \mathbb{R}^{M \times T}$ and its K DAGs. For child m , write

$$P_m = X_i(Pa(m), :) \in \mathbb{R}^{\tau_m \times T}, \quad \tau_m = |Pa(m)|.$$

Throughout the theorem's analysis, each child satisfies the exact within-batch relation $\mathbf{x}_{m,i} = \mathbf{w}_m P_m$, and P_m has full row rank. Thus the parent coefficients are identifiable. These assumptions concern the true trajectories. When invoking the per-child recovery statement, those parent trajectories are available exactly. Let $\tau = \max_m \tau_m$ over children and $l = \max_k |Root_k|$. The causal theorem concerns batches containing at least one child.

For an orthonormal basis $U \in \mathbb{R}^{T \times q}$ of a nonzero subspace, its incoherence is

$$\mu(U) = \frac{T}{q} \max_{1 \leq t \leq T} \|U^\top \mathbf{e}_t\|_2^2.$$

The bound μ applies to every parent row space. In the global-rank comparison it also bounds the row space of X_i .

A metric samples an integer number d of distinct time positions uniformly without replacement, independently of the data. Let $\Omega \in \{0, 1\}^{1 \times T}$ indicate those positions and let $D = \text{diag}(\Omega)$. For $q \geq 1$ and $0 < \varepsilon < 1$, define the sufficient integer budget

$$B(q, \varepsilon) = \min \{T, \lceil 8\mu q \log(q/\varepsilon) \rceil\}. \quad (1)$$

Full sampling applies whenever the uncapped bound reaches T . We set $B(0, \varepsilon) = 0$; exact membership in a zero-dimensional subspace means that the vector is zero.

LEMMA A.1 (RANK PRESERVATION UNDER UNIFORM SAMPLING). *Let $U \in \mathbb{R}^{T \times q}$ have orthonormal columns, $q \geq 1$, and $\mu(U) \leq \mu$. If $B(q, \varepsilon) \leq d \leq T$, then uniform sampling of d distinct time positions satisfies*

$$\Pr[\text{rank}(DU) = q] \geq 1 - \varepsilon.$$

Adding sampled positions preserves this rank.

PROOF. If $d = T$, every position is sampled and the conclusion is deterministic. Otherwise, the budget condition implies $d \geq 8\mu q \log(q/\varepsilon)$. Set $\mathbf{u}_t = U^\top \mathbf{e}_t$ and $A_t = \mathbf{u}_t \mathbf{u}_t^\top$. Then

$$\sum_{t=1}^T A_t = I_q, \quad 0 \preceq A_t \preceq \frac{\mu q}{T} I_q.$$

The sampled Gram matrix is

$$Y = (DU)^\top (DU) = \sum_{t: \Omega(t)=1} A_t, \quad \mathbb{E}[Y] = \frac{d}{T} I_q.$$

The lower-tail matrix Chernoff bound [2] also applies to this sum sampled without replacement. Specifically, the trace-exponential comparison of Gross and Nesme [1] bounds its moment-generating function by that of d independent uniform draws from the same finite indexed collection. Applying that comparison to $\text{tr exp}(-\theta Y)$, $\theta > 0$, transfers the Chernoff lower-tail bound. For $0 < \eta < 1$,

$$\Pr \left[\lambda_{\min}(Y) \leq (1 - \eta) \frac{d}{T} \right] \leq q \exp \left(-\frac{\eta^2 d}{2\mu q} \right).$$

Choosing $\eta = 1/2$ gives

$$\Pr \left[\lambda_{\min}(Y) \leq \frac{d}{2T} \right] \leq q \exp \left(-\frac{d}{8\mu q} \right) \leq \varepsilon.$$

On the complementary event, Y is positive definite and DU has full column rank. Each additional observation adds a positive semidefinite matrix A_t to the Gram matrix, so it cannot destroy this rank. $\square$

LEMMA A.2 (PARENT DIMENSION AND ROOT SPAN). *For every child m in DAG $\mathcal{G}_k$, under the exact linear relations and full-row-rank parent assumption,*

$$\tau_m = \text{rank}(C_m) \leq r_k \leq |\text{Root}_k| \leq l, \quad r_k = \text{rank}(X_i(\pi_k, :)).$$

Moreover, $\tau \leq \max_k r_k \leq r$, where $r = \text{rank}(X_i)$.

PROOF. Proceed in topological order within $\mathcal{G}_k$. Each root trajectory belongs to the span of the root trajectories. If every parent of a child belongs to that span, its exact linear relation puts the child in the same span. Induction therefore places every row of $X_i(\pi_k, :)$ in the root row span, giving $r_k \leq |\text{Root}_k|$.

The rows of P_m occur in $X_i(\pi_k, :)$ and are linearly independent. Appending their linear combination $\mathbf{x}_{m,i}$ does not increase rank, so $\text{rank}(C_m) = \text{rank}(P_m) = \tau_m$. This proves the first chain. Finally, $X_i(\pi_k, :)$ is a submatrix of X_i , so $r_k \leq r$; taking maxima proves the remaining claim. $\square$

PROOF OF THEOREM 4.2. (1) *Per-child reconstruction.* Fix a child m and its exact parent matrix P_m . Let $U_m \in \mathbb{R}^{T \times \tau_m}$ be an orthonormal basis of the parent row space. Since P_m has full row rank,

$$P_m = A_m U_m^\top$$

for an invertible $A_m \in \mathbb{R}^{\tau_m \times \tau_m}$. Lemma A.1 makes $D_m U_m$ full column rank with probability at least $1 - \delta$ whenever $d_m \geq B(\tau_m, \delta)$. On this event, $P_m D_m = A_m (D_m U_m)^\top$ has full row rank.

The true $\mathbf{w}_m$ has zero residual in Equation (7). Any other zero-residual minimizer $\mathbf{w}'$ would satisfy

$$(\mathbf{w}' - \mathbf{w}_m) P_m D_m = \mathbf{0},$$

which implies $\mathbf{w}' = \mathbf{w}_m$ by full row rank. Thus the sampled least-squares problem uniquely recovers the coefficients and $\mathbf{w}_m P_m = \mathbf{x}_{m,i}$ reconstructs the entire child trajectory.

From one DAG to K DAGs. Within $\mathcal{G}_k$, Lemma A.2 gives $\tau_m \leq |\text{Root}_k| \leq l$ and $\tau_m \leq \tau$. Consequently, the common per-child budget $c\mu\tau \log(l/\delta)$ is at least $8\mu\tau_m \log(\tau_m/\delta)$ when $c \geq 8$. It therefore suffices for any individual child in this DAG, given exact parent trajectories.

For K DAGs, use failure tolerance δ/K in each child's rank test:

$$8\mu\tau_m \log(K\tau_m/\delta) \leq c\mu\tau_m \log(Kl/\delta) \leq c\mu\tau \log(Kl/\delta).$$

After rounding upward and capping at T , either the local child budget or the theorem's uniform child budget therefore gives an individual success probability of at least $1 - \delta/K$, and hence at least $1 - \delta$. This establishes the first claim and supports the τ_m -dependent frequency in Equation (8) under the randomized sampling model. A union bound over one specified child from each DAG would also give joint success probability at least $1 - \delta$ for those selected children. It does not

establish simultaneous recovery of every child in all DAGs; the separate whole-batch extension below allocates failure probability over all metric recovery events.

(2) *Conditional comparison.* The rank chain in Lemma A.2 gives $\tau \leq r$. Under the additional condition $Kl \leq r$, monotonicity of the logarithm and positivity of $\log(Kl/\delta)$ yield

$$c\mu\tau \log(Kl/\delta) \leq c\mu r \log(Kl/\delta) \leq c\mu r \log(r/\delta).$$

Using the same incoherence upper bound and failure probability makes the right-hand side the corresponding sufficient budget for a known global row space. Upward rounding and capping at T preserve the comparison. The condition $Kl \leq r$ is sufficient, rather than a consequence of graph sparsity. The exact DAG relations also give $r \leq \sum_k |\text{Root}_k| \leq Kl$, so it requires linearly independent root trajectories and exactly l roots in every DAG. Without it, $\tau \leq r$ alone does not compare the two different logarithmic factors. These are sufficient-budget comparisons, not optimal sample lower bounds.

Limiting cases. For $K = 1$, the causal budget becomes $c\mu\tau \log(l/\delta)$. Under the comparison condition $l \leq r$, it is no larger than $c\mu r \log(r/\delta)$, although the improvement need not be strict. For $K = M$, every cluster contains one root and no child. The theorem, which assumes $\tau \geq 1$, does not apply in this case; there is no child coefficient to update, and all metrics use temporal SMC. Thus this limit removes the benefit of causal reconstruction rather than establishing a smaller total sampling requirement.

□

Separate whole-batch extension. To obtain simultaneous recovery, assume in addition that every root metric m has an exactly known historical matrix

$$\mathbf{H}_m = [\mathbf{x}_{m,i-W+1}^\top, \dots, \mathbf{x}_{m,i-1}^\top] \in \mathbb{R}^{T \times (W-1)},$$

with rank $q_m = \text{rank}(\mathbf{H}_m)$, its new vector belongs to $\text{col}(\mathbf{H}_m)$, and every nonzero historical column space has incoherence at most μ . The columns of $\mathbf{H}_m$ need not be linearly independent. With $c \geq 8$, a nonzero root space uses

$$d_m \geq \min \{T, \lceil c\mu q_m \log(q_m M/\delta) \rceil\}. \quad (2)$$

For a zero-dimensional root space, $d_m = 0$ suffices. Each child uses

$$d_m \geq \min \{T, \lceil c\mu\tau_m \log(KlM/\delta) \rceil\}. \quad (3)$$

For each metric, choose its positions uniformly without replacement and independently of the data. Dependence between different metrics' sample sets is allowed.

For a nonzero root space, Lemma A.1 with $\varepsilon = \delta/M$ preserves the rank of the sampled historical matrix. To see why this suffices even with dependent historical columns, write $\mathbf{H}_m = \mathbf{Q}_m \mathbf{R}_m$, where $\mathbf{Q}_m$ is an orthonormal basis of its q_m -dimensional column space and $\mathbf{R}_m$ has full row rank. Full column rank of $\mathbf{D}_m \mathbf{Q}_m$ implies

$$\ker(\mathbf{D}_m \mathbf{H}_m) = \ker(\mathbf{H}_m).$$

Since the true new vector is in $\text{col}(\mathbf{H}_m)$, the sampled least-squares system is consistent. Its pseudoinverse solution therefore satisfies

$$\mathbf{H}_m (\mathbf{D}_m \mathbf{H}_m)^\dagger \mathbf{D}_m \mathbf{x}_{m,i}^\top = \mathbf{x}_{m,i}^\top.$$

Indeed, its coefficient error lies in $\ker(\mathbf{D}_m \mathbf{H}_m) = \ker(\mathbf{H}_m)$. The historical coefficients may be nonunique, but the reconstructed trajectory is exact. A root with $q_m = 0$ is zero by the subspace membership assumption.

For each child, $\tau_m \leq Kl$ makes its budget in Equation (3) at least $B(\tau_m, \delta/M)$, so its sampled parent matrix preserves rank with failure probability at most δ/M . A union bound over the at most M root and child rank-preservation events makes all of them hold with probability at least $1 - \delta$.

No independence between these events is needed. On their intersection, first reconstruct the roots as above, then proceed in topological order. Each child receives exact parent trajectories, and the coefficient argument in the theorem's proof recovers that child exactly. This reconstructs the entire batch.

Relation to the implementation. The theorem and its extension analyze random time-position sampling. The periodic base schedule in the implementation is not covered by this probability guarantee. Additional observations preserve the sampled rank, provided the stated random base sample set is already present. Exact recovery also requires the stated subspace membership and exact linear relations; approximate parent reconstruction or model error requires a separate stability analysis.

For roots, the general rank inequality $q_m \leq \text{rank}([H_m, \mathbf{x}_{m,i}^\top]) \leq q_m + 1$ supports a conservative budget proxy. It does not provide a basis for an unseen direction. The whole-batch extension above uses membership in the existing historical span, while the implementation's $\tilde{r}_m = q_m + 1$ is only a sample-allocation proxy.

B Anomaly-Model Learning and Updating

This appendix derives the approximate EM updates used by the Anomaly Sampler and Model Updater (Sections 4.3 and 4.6). The SCS fixes the allowed source metrics $m' \in Pa(m) \cup \{m\}$; only their propagation strengths and decay rates are learned. We use the exponential kernel $g_{mm'}(v) = a_{mm'}\beta_{mm'}e^{-\beta_{mm'}v}$ for $v > 0$ and the conditional event rate in Equation (9).

B.1 Likelihood and Event-Source Probabilities

Let h denote the final observed time step: $h = WT$ during offline learning and $h = iT$ after online batch i . The modeled history begins at time 0, and $E_m(h+1)$ includes all detected events of metric m through time h . For the continuous-time Hawkes model evaluated at these event times, the log-likelihood is

$$\ell(\theta) = \sum_{m=1}^M \left[\sum_{s \in E_m(h+1)} \log \lambda_m(s) - \int_0^h \lambda_m(t) dt \right]. \quad (4)$$

The event terms reward rates that explain the detected anomalies, while the integrated rate also accounts for time without events. For the exponential kernel, the integrated rate is

$$\int_0^h \lambda_m(t) dt = u_m h + \sum_{m' \in Pa(m) \cup \{m\}} a_{mm'} B_{mm'}(\beta_{mm'}; h), \quad (5)$$

where the finite-horizon contribution of source metric m' is

$$B_{mm'}(\beta; h) = \sum_{v \in E_{m'}(h+1)} \left(1 - e^{-\beta(h-v)} \right). \quad (6)$$

Following the branching interpretation of Hawkes processes [3], each target event has an unobserved source: spontaneous occurrence or triggering by an earlier self or parent event. With the parameters fixed at their current values, the E-step assigns the source probabilities

$$p_{ss}^m = \frac{u_m}{\lambda_m(s)}, \quad p_{sv}^{mm'} = \frac{g_{mm'}(s-v)}{\lambda_m(s)}, \quad v \in E_{m'}(s). \quad (7)$$

The allowed sources satisfy $m' \in Pa(m) \cup \{m\}$, and $E_{m'}(s)$ contains events strictly before s . These probabilities sum to one for each target event when $\lambda_m(s) > 0$.

B.2 Offline Parameter Updates

At each offline iteration, we recompute the source probabilities and the following statistics over the training events:

$$S_{u_m} = \sum_{s \in E_m(h+1)} p_{ss}^m, \quad S_{a_{mm'}} = \sum_{s \in E_m(h+1)} \sum_{v \in E_{m'}(s)} p_{sv}^{mm'}. \quad (8)$$

$$S_{\beta_{mm'}} = \sum_{s \in E_m(h+1)} \sum_{v \in E_{m'}(s)} p_{sv}^{mm'} (s - v). \quad (9)$$

Here S_{u_m} is the expected number of spontaneous events, $S_{a_{mm'}}$ is the expected number triggered by metric m' , and $S_{\beta_{mm'}}$ is their probability-weighted total triggering delay.

Holding these probabilities fixed, the expected complete-data log-likelihood separates into a background term $Q_{u_m} = S_{u_m} \log u_m - u_m h$ and, for each allowed source–target pair,

$$Q_{mm'} = S_{a_{mm'}} (\log a_{mm'} + \log \beta_{mm'}) - \beta_{mm'} S_{\beta_{mm'}} - a_{mm'} B_{mm'}(\beta_{mm'}; h). \quad (10)$$

Terms independent of the updated parameters are omitted. The background update is $u_m = S_{u_m}/h$. For an identifiable source–target pair, the exact finite-horizon stationary conditions are

$$a_{mm'} = \frac{S_{a_{mm'}}}{B_{mm'}(\beta_{mm'}; h)}, \quad (11)$$

$$\frac{S_{a_{mm'}}}{\beta_{mm'}} - S_{\beta_{mm'}} - a_{mm'} \sum_{v \in E_{m'}(h+1)} (h - v) e^{-\beta_{mm'}(h-v)} = 0. \quad (12)$$

Thus, exact finite-horizon updates couple the propagation strength and decay rate.

CauSample uses a boundary approximation to obtain explicit updates. Writing $n_{m'} = |E_{m'}(h+1)|$, we replace $B_{mm'}(\beta; h)$ by $n_{m'}$ and omit its derivative with respect to β . The resulting approximate M-step is

$$u_m = \frac{S_{u_m}}{h}, \quad a_{mm'} = \frac{S_{a_{mm'}}}{n_{m'}}, \quad \beta_{mm'} = \frac{S_{a_{mm'}}}{S_{\beta_{mm'}}}. \quad (13)$$

These ratios apply when the denominators are positive and the event statistics identify the corresponding parameters. The approximation requires both the omitted kernel tails $\sum_v e^{-\beta_{mm'}(h-v)}$ and their derivative contributions to be small relative to the retained terms. A long observation horizon alone does not ensure this condition when many source events occur near its end. Consequently, these are approximate EM updates, rather than exact M-steps for the finite-horizon objective in Equation (10). We repeat the offline E- and M-steps with $h = WT$ until the maximum parameter change falls below ϵ .

B.3 Incremental Updates after Each Batch

We initialize the cumulative statistics from the final offline E-step. After batch i , each newly sampled anomaly at time s in metric m undergoes one E-step followed by one approximate M-step. Its source probabilities are computed using the current parameters and only events strictly before s , then added to the cumulative statistics:

$$S_{u_m} \leftarrow S_{u_m} + p_{ss}^m, \quad S_{a_{mm'}} \leftarrow S_{a_{mm'}} + \sum_{v \in E_{m'}(s)} p_{sv}^{mm'}, \quad (14)$$

$$S_{\beta_{mm'}} \leftarrow S_{\beta_{mm'}} + \sum_{v \in E_{m'}(s)} p_{sv}^{mm'} (s - v). \quad (15)$$

For $m' \in Pa(m) \cup \{m\}$, we update the corresponding parameters using Equation (13) with $h = iT$ and $n_{m'} = |E_{m'}(iT + 1)|$. Both quantities include the initial training period and all subsequent

batches. They remain fixed while processing the new events from the current batch; each parameter update precedes the E-step for the next event. The resulting parameters are used for predictions in the next batch.

Past source probabilities remain fixed, including contributions from batches that have left the sliding window. This cumulative update avoids repeated EM iterations over old target events, although each new target event still uses earlier parent and self events to compute its source probabilities. It does not claim exact maximization or monotonic improvement of the full-history likelihood. If a batch contains no sampled anomalies, no parameter update is performed; its elapsed time is included when the next event triggers an update. Within-batch predictions incorporate new events with fixed parameters, whereas parameter learning takes place after the batch ends.

References

- [1] David Gross and Vincent Nesme. 2010. Note on sampling without replacing from a finite collection of matrices. arXiv:1001.2738 <https://arxiv.org/abs/1001.2738>
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- [3] Alejandro Veen and Frederic P. Schoenberg. 2008. Estimation of Space–Time Branching Process Models in Seismology Using an EM-Type Algorithm. *J. Amer. Statist. Assoc.* 103, 482 (2008), 614–624. doi:10.1198/016214508000000148